

Course ID: SW 302

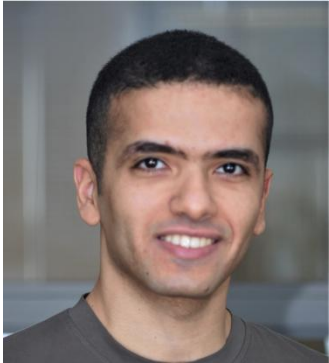
User Interface (UI) Development

Lecture#1

Dr. Mohamed Sami Rakha

Intro & Course Outline

About me...



- **Dr. Mohamed Sami Rakha**
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Lecture Outline

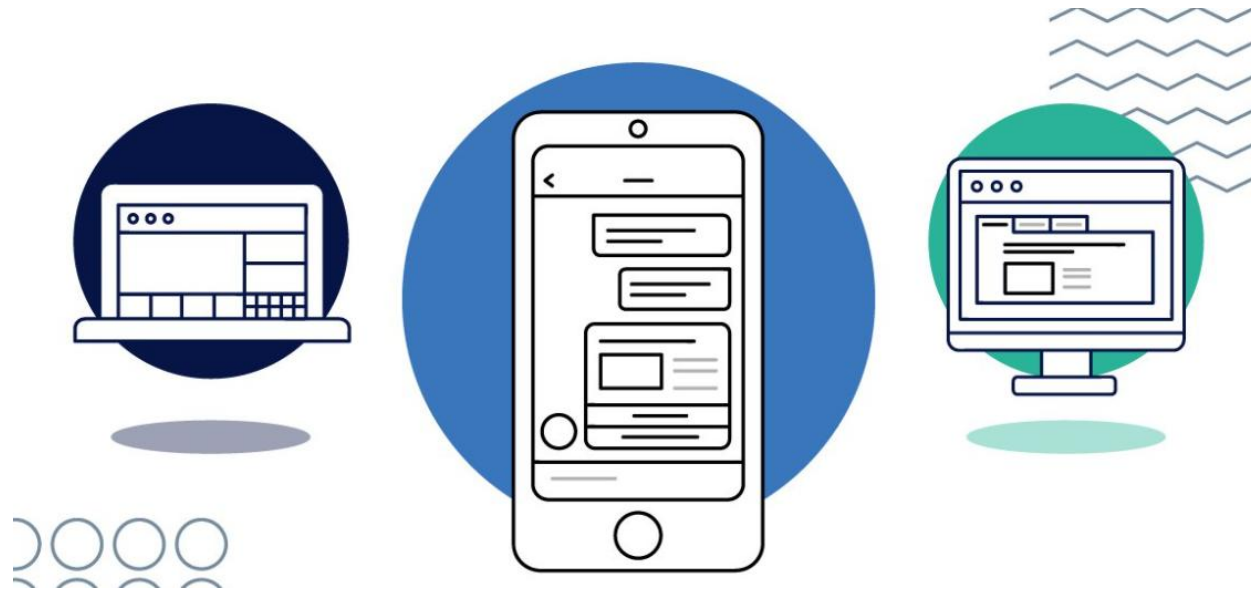
- What is UI Development?
- Covered Topics
- Course Administration
- Course LOs

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What is UI?

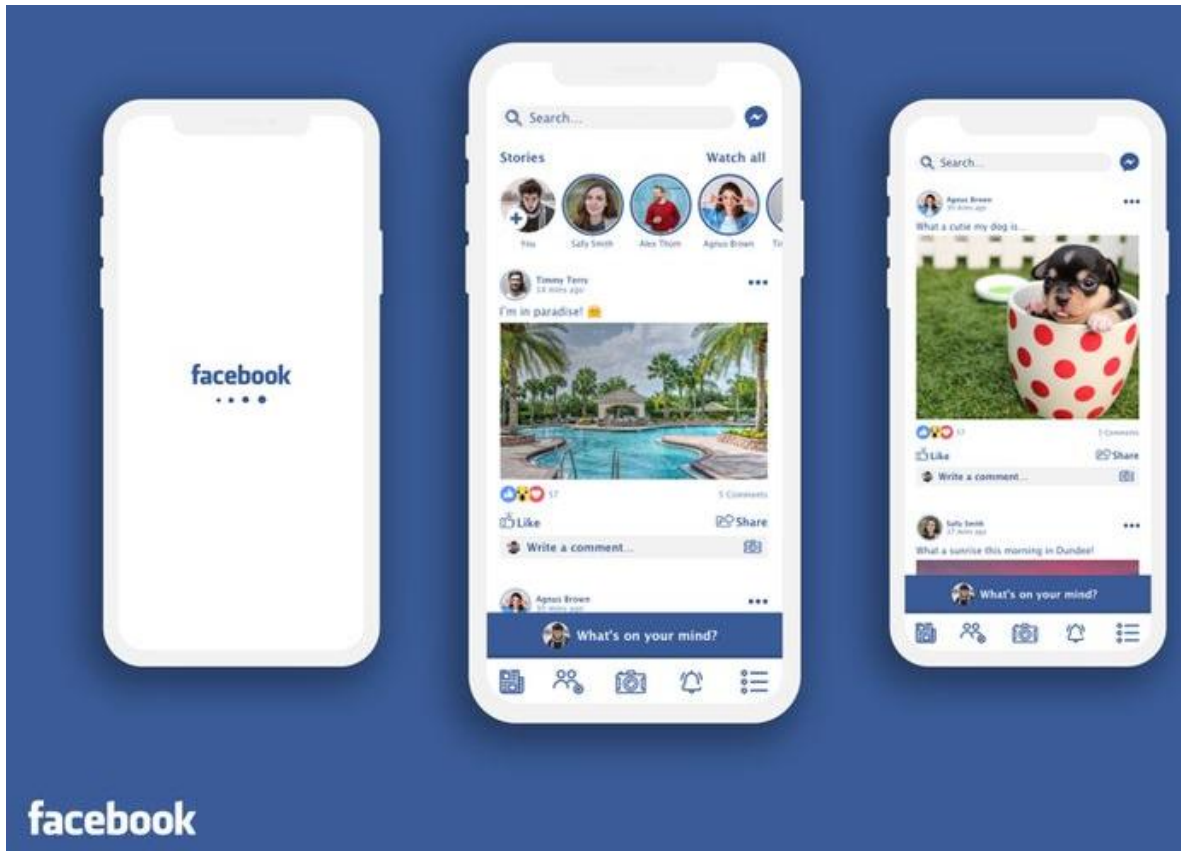
What a UI?

A **User Interface (UI)** is the point of interaction between a user and a digital **product** (website, mobile app, software, or device). It encompasses everything the user sees and interacts with, including **buttons, menus, icons, text fields, images, layouts, and colors.**

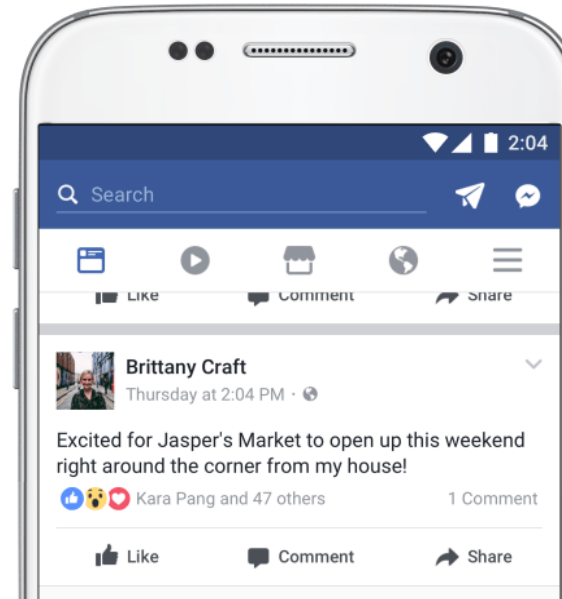


UI Examples

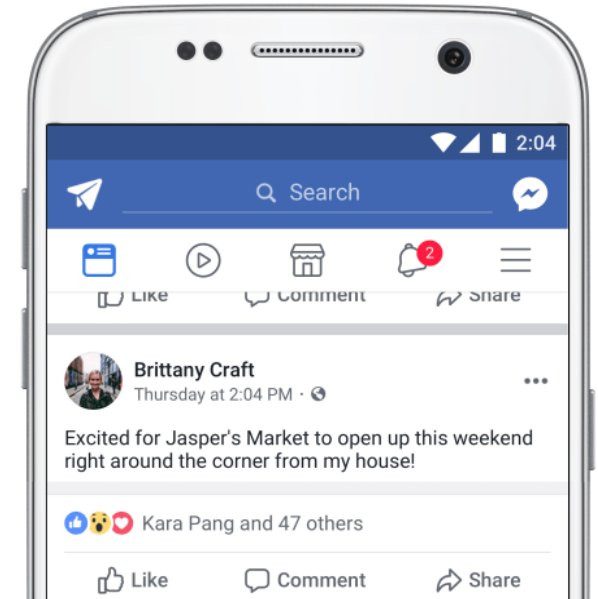
Facebook Mobile Apps UIs – fit to the screen- responsive !



Before

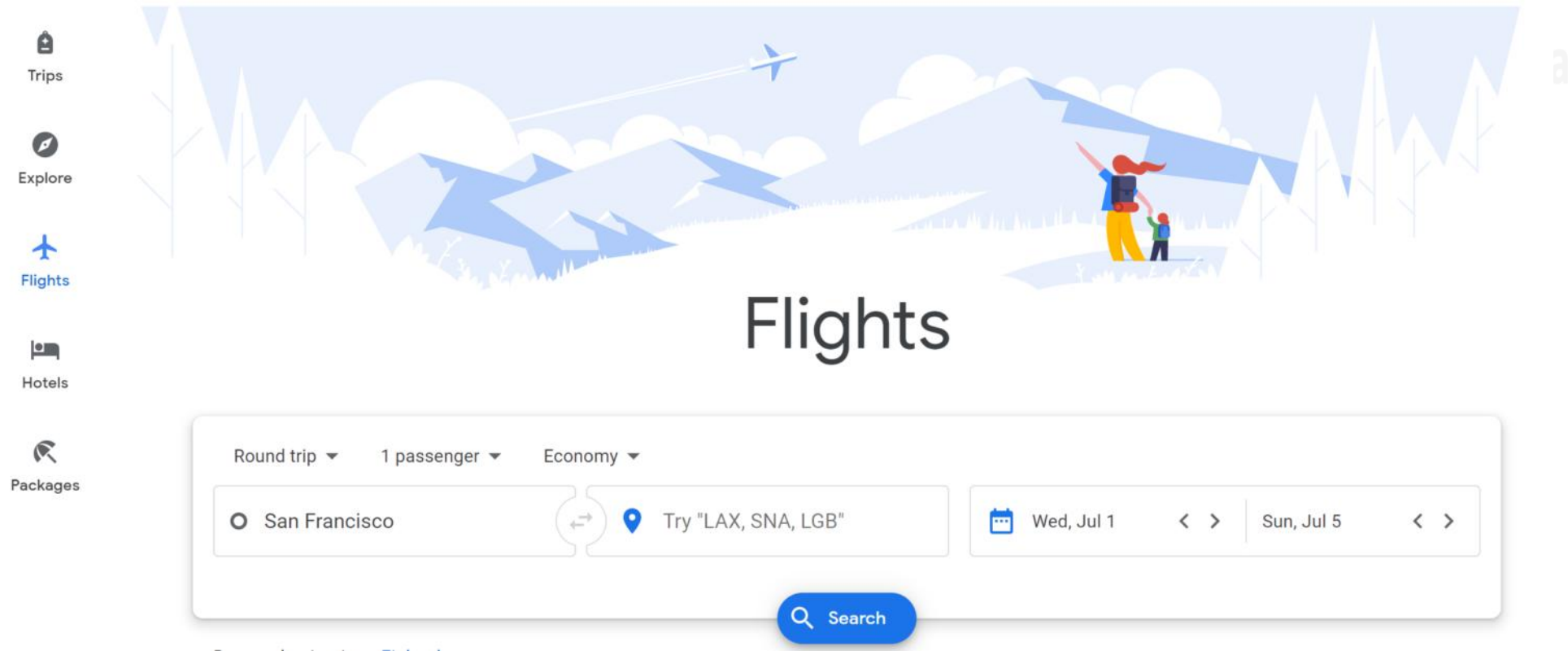


After



UI Examples

UI from Google Flights – simple!



The image shows a UI mockup for Google Flights. On the left is a vertical sidebar with icons and labels for 'Trips', 'Explore', 'Flights' (highlighted in blue), 'Hotels', and 'Packages'. The main area features a large, stylized illustration of a landscape with mountains, trees, and a person with a backpack. Below the illustration, the word 'Flights' is displayed in a large, dark font. At the bottom is a search bar with several fields: 'Round trip' with a dropdown arrow, '1 passenger' with a dropdown arrow, 'Economy' with a dropdown arrow, a departure field containing 'San Francisco', a swap button, a destination field containing 'Try "LAX, SNA, LGB"', a date field showing 'Wed, Jul 1' and 'Sun, Jul 5' with navigation arrows, and a blue 'Search' button with a magnifying glass icon.

Trips

Explore

Flights

Hotels

Packages

Flights

Round trip ▾ 1 passenger ▾ Economy ▾

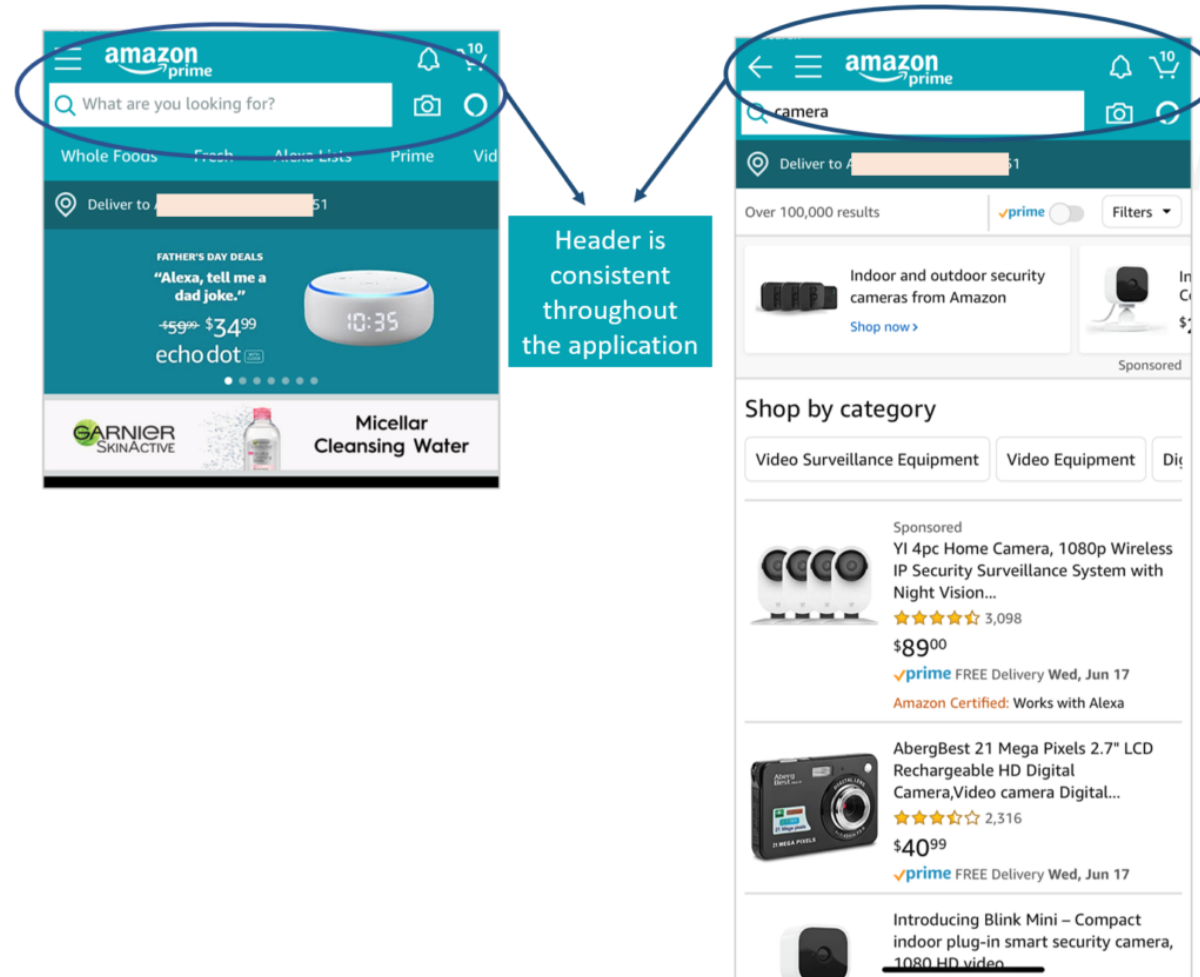
○ San Francisco ↔ Try "LAX, SNA, LGB"

Wed, Jul 1 < > Sun, Jul 5 < >

Search

UI Examples

UI from Amazon – consistent



UI Examples

ATM UI – easy to use custom!



UI Examples

Car Dashboard!

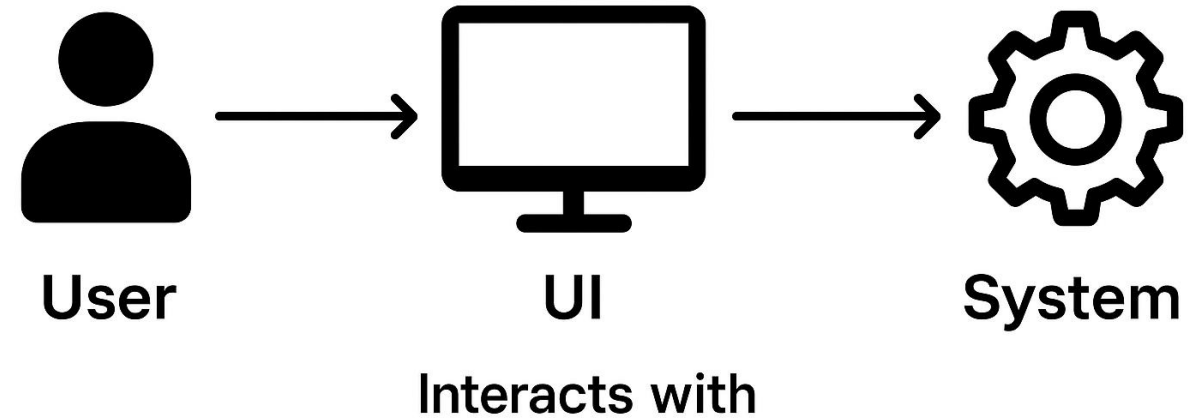


Rakha

UI as a Bridge Between User and System

The relationship between the **User**, the **UI**, and the **System/Software**

- The **User** interacts with the **UI** (what they see and click).
- The **UI** acts as the bridge between the **User** and the **System**.
- The **System** processes actions and sends results back via the **UI**.



Ui vs UX?

UI vs User Experience (UI vs UX)

User Interface (UI):

- The **look and feel** of the product
- Visual elements (colors, fonts, icons, buttons, layout)
- Focus: **How it looks**

User Experience (UX):

- The **overall experience** of using the product
- Ease of navigation, performance, accessibility, satisfaction
- Focus: **How it works & feels**



UI vs UX

Left: A visually beautiful chair (UI)

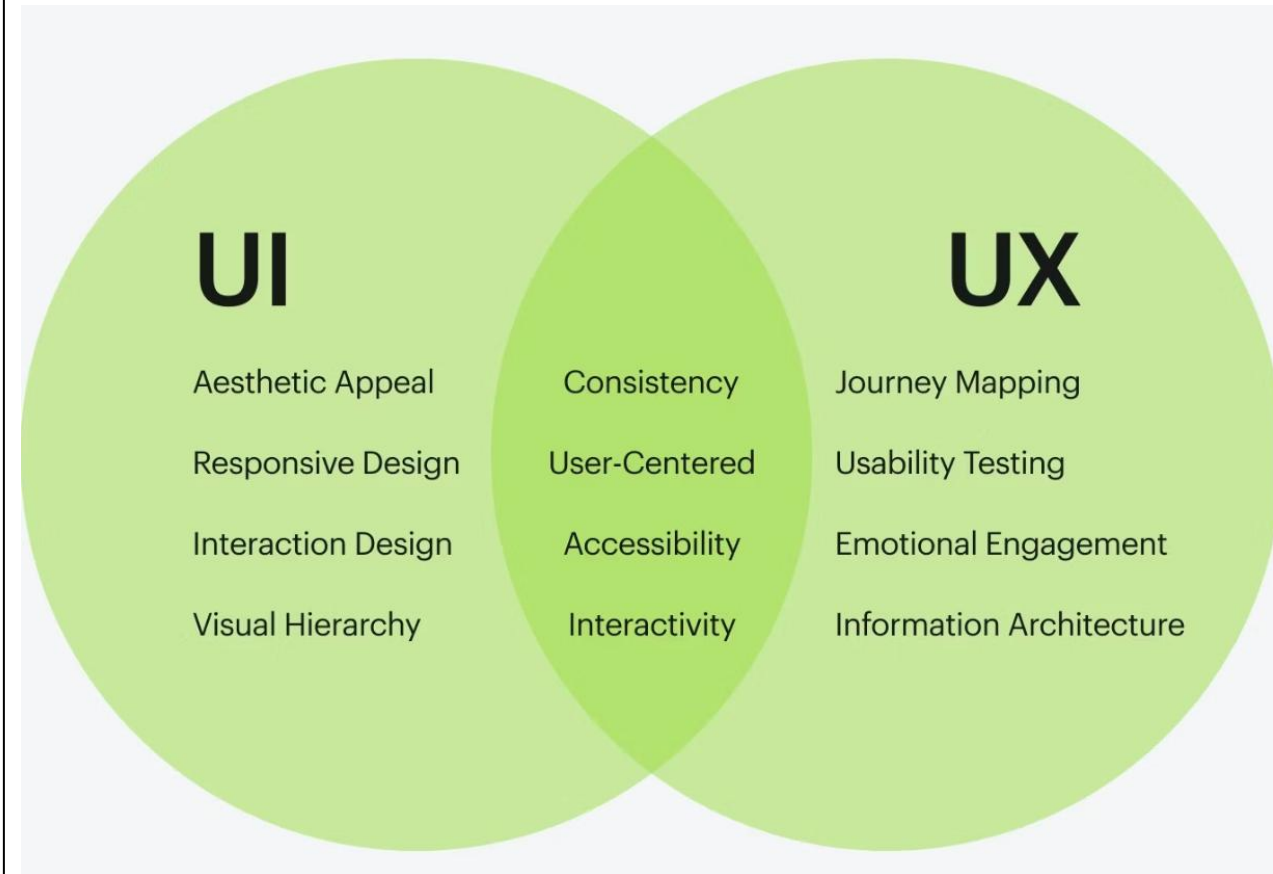


Right: A comfortable chair you actually enjoy sitting in (UX)



UI vs UX: Different Disciplines that Work in Harmony

- **UI design and UX design** involve very different skill sets, but they are integral to each other's success.
- **A beautiful design** can't save an interface that's confusing to navigate, and a brilliant, perfectly-appropriate user experience can be sunk by bad visual interface design that makes using the app unpleasant.
- **Both UI and UX designs** need to be flawlessly executed and perfectly aligned with pre-existing user expectations to create an excellent user interface/experience.



So... what is UI Development?

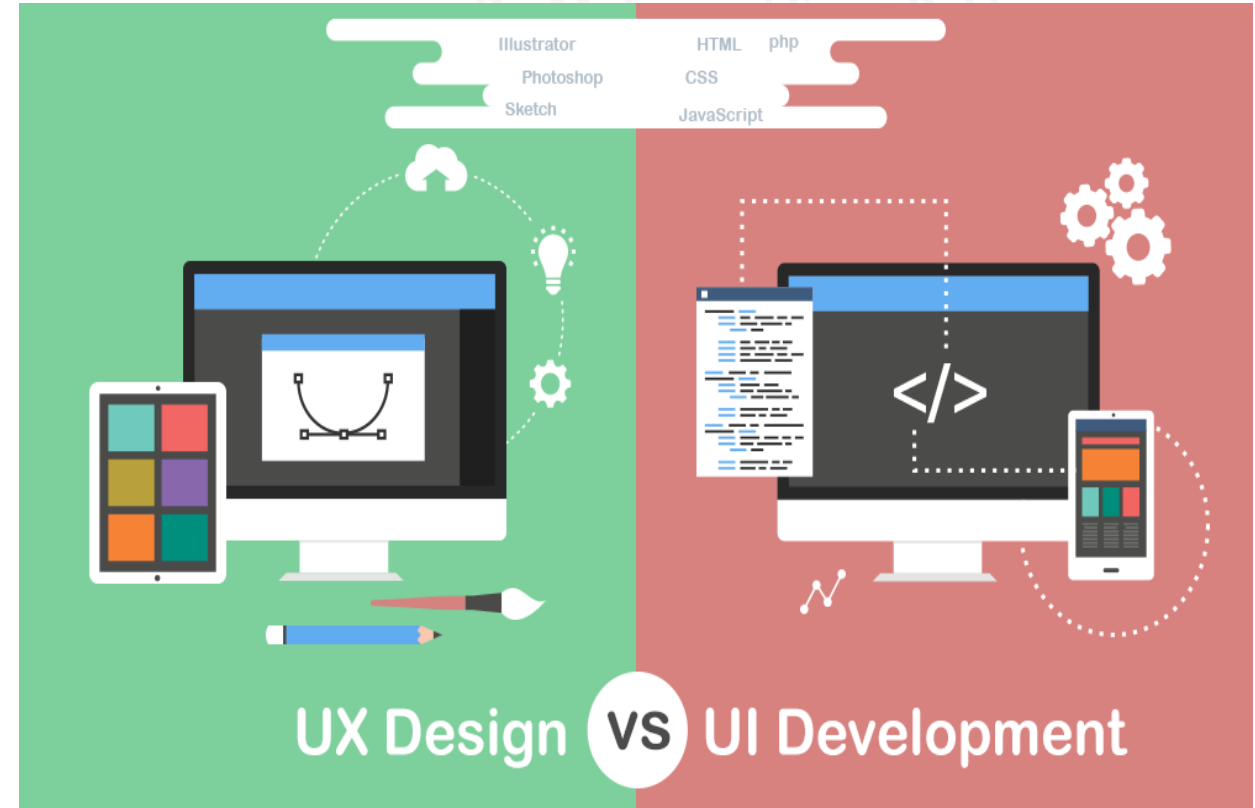
What a UI?

UI Development is the process of building the interactive parts of software applications that users see and interact with.

- Includes web apps, mobile apps, desktop software, smart devices, ATMs, car dashboards

Uses different technologies depending on the platform:

- **Web apps** → HTML, CSS, JavaScript, React
- **Mobile apps** → Java (Android), Flutter/React Native
- **Desktop apps** → WPF/.NET, JavaFX, Qt



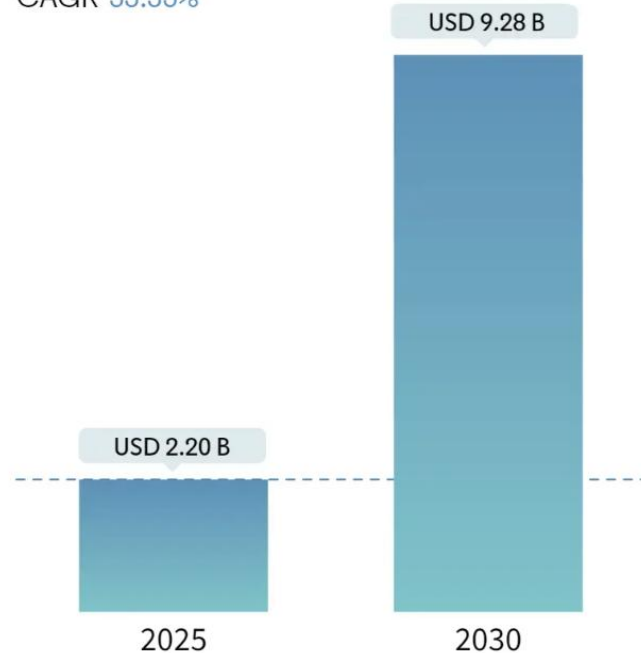
How big is UI Development market?

The UI/UX Market size is estimated at USD 2.20 billion in 2025, and is expected to reach USD 9.28 billion by 2030

Source: <https://www.mordorintelligence.com/industry-reports/ui-ux-market>

UI/UX Market

Market Size in USD Billion
CAGR 33.35%



Source : Mordor Intelligence



Global UI and UX Design Software Market Revenue (M USD) in 2025



MAIA



+ 22.25%

Global Market CAGR
2025-2030



ACCELERATING

Growth Momentum

Source:
www.maiaresearch.com

Covered Topics

- Introduction
- HTML/CSS
- Document Object Model
- Modern JavaScript
- React
- Routing & SPAs
- State Management
- Testing & DevOps Basics
- DevOps for Frontend

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Course Administration

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- **Course Grades:**

- Project: 20
- Midterm Exam: 20
- Lab Assignments & Quizzes & Others: 20

(15 grades for 3x Assignments + 5 grades for Lab Participation)

- Final Exam: 40
- **Four lab assignments**
- **Exam System**
 - MCQs and others

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Course Learning Outcomes (CLO)

- Design responsive web pages using HTML and CSS.
- Develop interactive web applications using JavaScript/React.
- Implement asynchronous data fetching using React to improve user experience.
- Utilize DOM manipulation and state management.
- Evaluate and troubleshoot web applications for cross-browser compatibility, web accessibility, and responsiveness.

Policy on the Use of AI / Academic Integrity

Use of AI is permitted (unless mentioned otherwise), **but:**

- Students must properly document how AI was used.
- The work will still be subject to evaluation of the student's own understanding through discussion or examination.

Academic Integrity and Understanding

- All submitted work must reflect the student's own effort and understanding.
- Instructors / TA may conduct an oral or written discussion with students about their submitted work.
- **If a student cannot adequately explain or demonstrate understanding of their submission, the work will be considered non-authentic and treated as cheating.**

Consequences of Violation:

- **First Offense:**
 - The student will receive a zero grade on the concerned assignment/quiz/exam.
 - The incident will be flagged as cheating.
- **Second Offense:**
 - The case will be referred to the Student Conduct Committee.
 - Penalties may include failing the course and additional disciplinary action as per university regulations.

Textbooks / References

- **JavaScript for Web Warriors, 7th edition, Book by Patrick Carey and Sasha Vodnik**
- **Additional references will be provided during the course.**

Thank You...